

Laboratory 10:

Introduction to CUDA

Introduction:

To use CUDA, your computer should have Nvidia GPU installed. If you don't have one, you can use torus server:

ssh surname.name.id_number@torus.uck.pk.edu.pl – default password is stdpkpass (change it!)

Steps:

1. Create a working directory (e.g., *Cuda*).
2. Compile (nvcc) and run the [program](#) that shows information about the available CUDA device. Familiarize yourself with the device parameters. **Remeber that for nvcc to work you need .cu file (not .c or .cpp!)**
3. Familiarize yourself with two levels of parallelization - [through blocks](#) and [through threads](#).
4. Compare and analyze the division of work in two programs - with the [vector size divided by the number of blocks](#) and [with the size indivisible by the number of blocks](#).
5. Modify the last of the programs with the procedure of adding vectors on the CPU, add time measurement procedures and measure the time for the largest possible size of the vectors. Three times should be measured - the first as the CPU computing time, the second as the time of the GPU calculations without copying the data from memory, and the third as the duration of the entire operation on the graphics card. The results should be presented on a comparative graph for different sizes of vectors.
6. Analyze the [program](#) that adds parts of the vector with respect to the defined radius on the right and left. Check how thread synchronization and shared memory usage works.



7. Develop a sequential program to calculate Euler's gamma constant:

$$g = \sum_{k=1}^n \left(\frac{1}{k} \right) - \ln(n)$$

8. Run the program sequentially; check the correctness of the result obtained for different **n**.
9. Parallelize written procedure using CUDA.
10. Add timing routines and compare execution times for sequential and parallel versions (graphs).